AIR UNIVERSITY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

EXPERIMENT NO 4

Lab Title Graphs

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Objective:

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____



Department of Electrical and Computer Engineering

Signals and Systems Lab

Lab Report: 4

Title: Graphs

Name: Ali Abbas Khan

Roll Number: 230661

Class: BCE-5B

Submitted To: Engr. Maryam
Date: October 15, 2025

1. Plot the following function in MATLAB.

$x(t) = A * \sin((2 * pi * f * t) + \theta)$

where, frequency f = 100 Hz, t is from 0 to 10 seconds and angle = 4 and A is the amplitude with
the value 4.

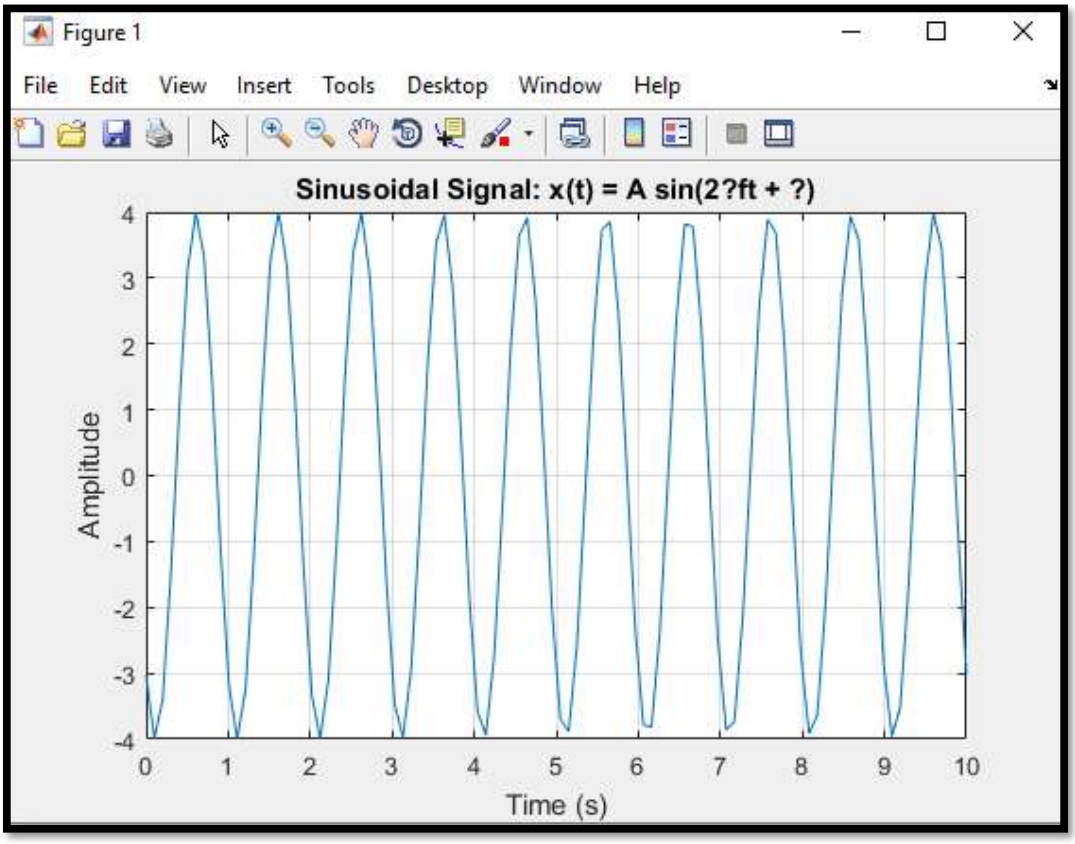
Code:

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T1.m

```
1 -    f = 100;
2 -    A = 4;
3 -    theta = 4;
4 -    t = linspace(0,10);
5 -    x = A * sin(2 * pi * f * t + theta);
6 -    figure;
7 -    plot(t,x);
8 -    xlabel('Time (s)');
9 -    ylabel('Amplitude');
10 -   title('Sinusoidal Signal: x(t) = A sin(2πft + θ)');
11 -   grid on;
```

Output:



2. Plot the given function:

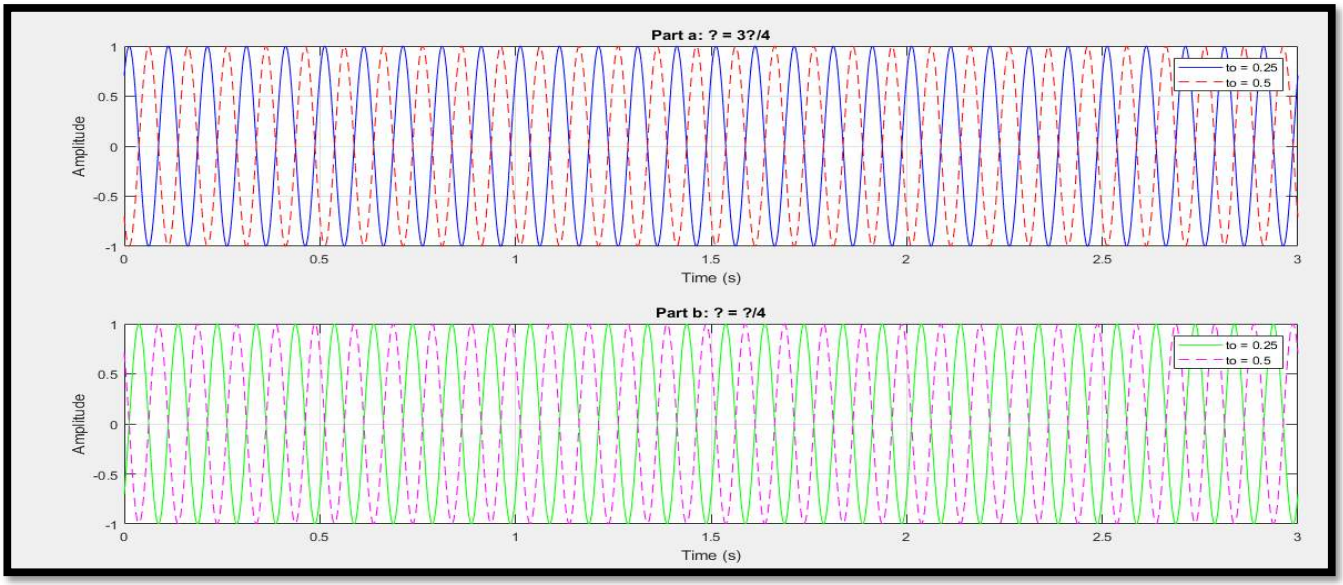
$x(t) = \cos(2 * pi * (t - to) + \theta)$

where:
a. frequency f = 10 Hz , t is from 0 to 3 seconds and angle θ is 3π/4. Apply time shift to this signal
toward right by time to. Take time shift to = 0.25and0.5 seconds.
b. frequency f = 10 Hz , t is from 0 to 3 seconds and angle is π/4. Apply time shift to this signal
toward right by time to. Take time shift to = 0.25and0.5 seconds. Use Subplot()

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T1.m | T2.m*

```
f = 10;
t = linspace(0, 3, 3001);
to1 = 0.25;
to2 = 0.5;
theta_a = 3*pi/4;
x_a1 = cos(2*pi*f*(t - to1) + theta_a);
x_a2 = cos(2*pi*f*(t - to2) + theta_a);
theta_b = pi/4;
x_b1 = cos(2*pi*f*(t - to1) + theta_b);
x_b2 = cos(2*pi*f*(t - to2) + theta_b);
figure;
subplot(2,1,1);
plot(t, x_a1, 'b', t, x_a2, 'r--');
title('Part a: θ = 3π/4');
xlabel('Time (s)');
ylabel('Amplitude');
legend('to = 0.25', 'to = 0.5');
grid on;
subplot(2,1,2);
plot(t, x_b1, 'g', t, x_b2, 'm--');
title('Part b: θ = π/4');
xlabel('Time (s)');
ylabel('Amplitude');
legend('to = 0.25', 'to = 0.5');
grid on;
```



3. Use MATLAB to plot the following equations in one figure :
t^2
t^3
e^t

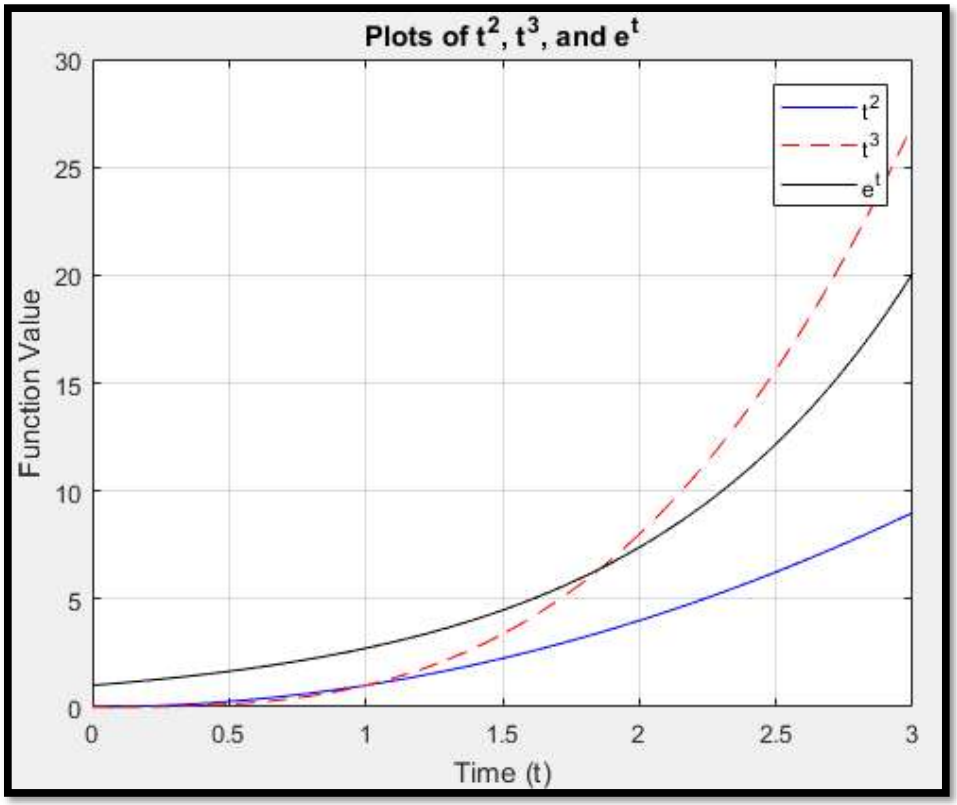
Editor - C:\Users\Admin\OneDrive\Desktop\SNS Lab\Matlab Files\L4\T3.m

T1.m | T2.m | T3.m

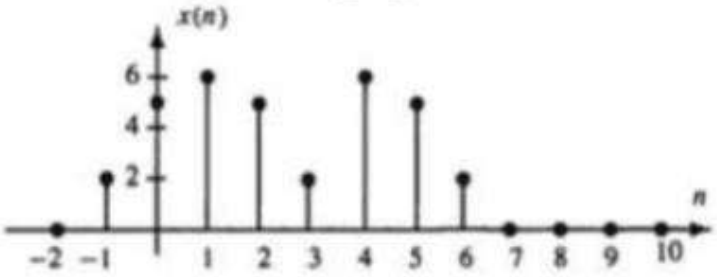
```
t = linspace(0, 3, 3001);

y1 = t.^2;
y2 = t.^3;
y3 = exp(t);

figure;
plot(t, y1, 'b', t, y2, 'r--', t, y3, 'k-');
title('Plots of t^2, t^3, and e^t');
xlabel('Time (t)');
ylabel('Function Value');
legend('t^2', 't^3', 'e^t');
grid on;
```



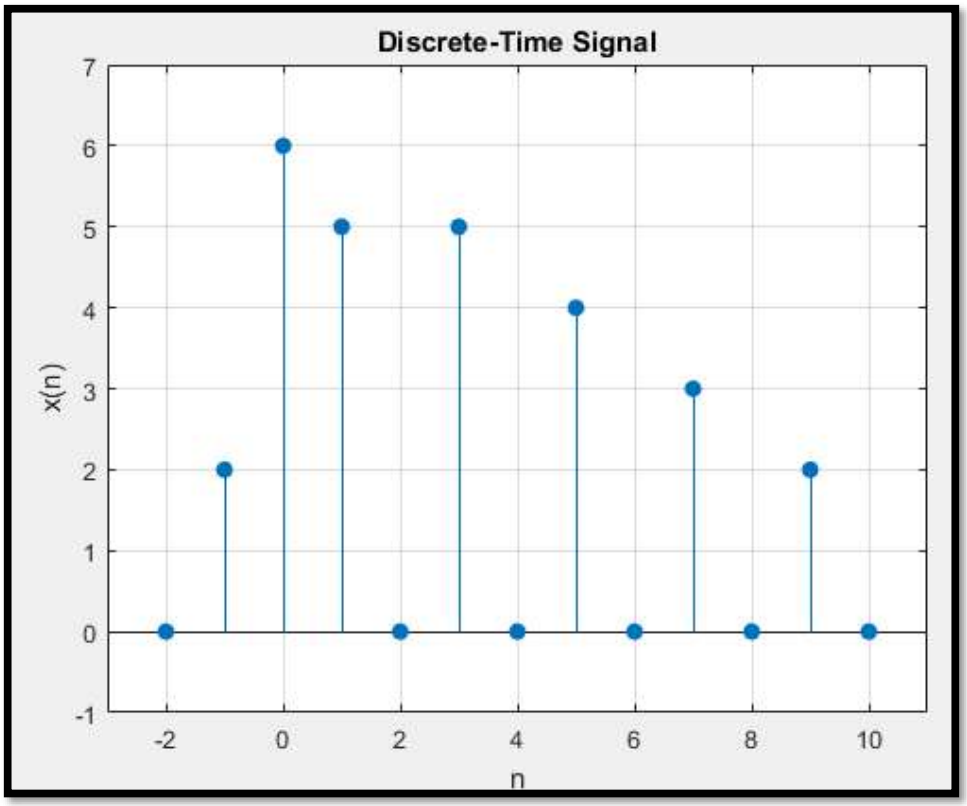
4. Plot The Following Signal In MATLAB



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T1.m | T2.m | T3.m | T4.m

```
1 -    n = -2:10;
2 -    x = [0 2 6 5 0 5 0 4 0 3 0 2 0];
3
4 -    stem(n, x, 'filled');
5 -    xlabel('n');
6 -    ylabel('x(n)');
7 -    title('Discrete-Time Signal');
8 -    grid on;
9 -    axis([-3 11 -1 7]);
```



5. Generate bar graph for the following data in MATLAB.

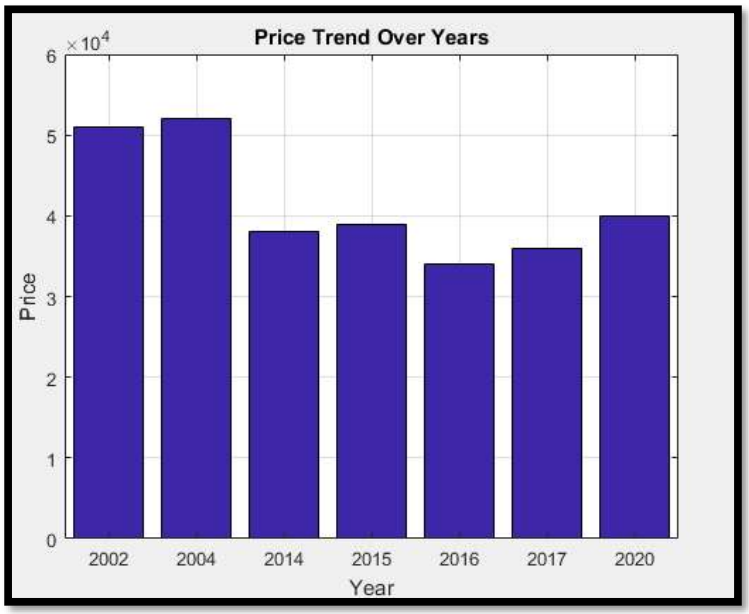
Year	Price
2002	51000
2004	52000
2014	38000
2015	38,850
2016	34000
2017	36000
2020	40000

(Hint: Year should be on the x-axis and each bar should have a categorical tick label of the respective year)

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T1.m | T2.m | T3.m | T4.m | T5.m

```
1 -    Year = categorical({'2002','2004','2014','2015','2016','2017','2020'});
2 -    Price = [51000 52000 38000 38850 34000 36000 40000];
3
4 -    bar(Year, Price);
5 -    xlabel('Year');
6 -    ylabel('Price');
7 -    title('Price Trend Over Years');
8 -    grid on;
```



Conclusion:

This lab provided a hands-on exploration of signal visualization and MATLAB fundamentals, guiding us through plotting continuous and discrete-time signals, applying time and phase shifts, and representing real-world data using bar graphs. By working with mathematical expressions and structured datasets, we reinforced key concepts in signal processing and data presentation—building both technical fluency and analytical clarity.